

Day 55 - Wireless Fundamentals

Wednesday, September 9, 2026

8:38 PM

Wireless Fundamentals Info

The standard that is used for Wireless LANs are defined IEEE 802.11.

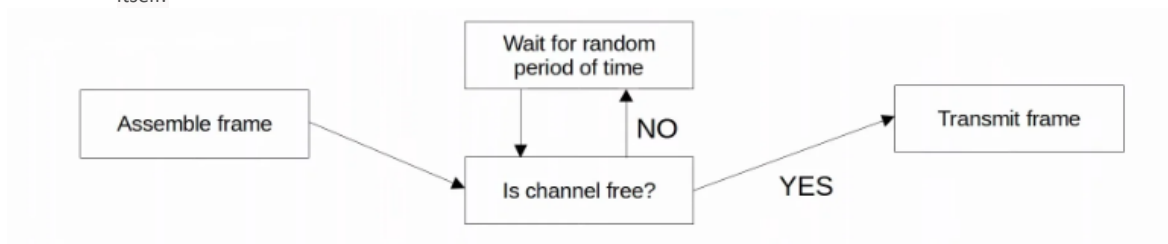
- The **Wi-Fi** is a trademark of the **Wi-Fi Alliance**, not directly connected to the previously mentioned **802.11**.
- The **Wi-Fi Alliance** tests and certifies equipment for **802.11** standards compliance interoperability with other devices.

Wireless Networks Problems

There are some problems with wireless networks that need to be dealt with.

1. Privacy

- All devices within range receive all frames, like devices connected to an Ethernet hub.
 - This can lead to privacy problems within the LAN.
 - **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)** is used to facilitate *half-duplex* communications
 - When using **CSMA/CA**, a device will wait for other devices to stop transmitting before it transmits data itself.



CSMA/CA Demonstration

CSMA/CD VS CSMA/CA

- **CSMA/CD** is used in *wired* networks to detect and recover from collisions.
- **CSMA/CA** is used in *wireless* networks to avoid collisions.

2. Regulations

Wireless communications are regulated by various international and national bodies.

3. Signal Coverage

Wireless signal coverage area must be considered.

- Signal range.
- Signal **absorption, reflection, refraction, diffraction**, and **scattering**.
 1. **Absorption**
 - This happens when a wireless signal passes through a material and is converted into heat, weakening the original signal.
 2. **Reflection**
 - This happens a signal bounces off of a material, of example metal.
 - An example of **reflection** happens when you find that your Wi-Fi reception to be poor inside of elevators. This happens because the signal bounces of the metal and very little penetrates into the elevator.
 3. **Refraction**
 - This happens when a wave is bent when entering a medium where the signal travels at a different speed.
 4. **Diffraction**
 - This happens when a wave encounters an obstacle and travels around it.
 5. **Scattering**
 - This happens when a material causes a signal to scatter in all directions.

4. Interference

- Other devices using the same wireless channels can cause interference.
 - For example, a wireless LAN in your neighbor's house/apartment.

Radio Frequency

Wireless signals are basically electromagnetic fields which propagate out as waves.

- Electromagnetic waves can measured in multiple ways for example:

- Electromagnetic waves can be measured in multiple ways for example:

1. Amplitude

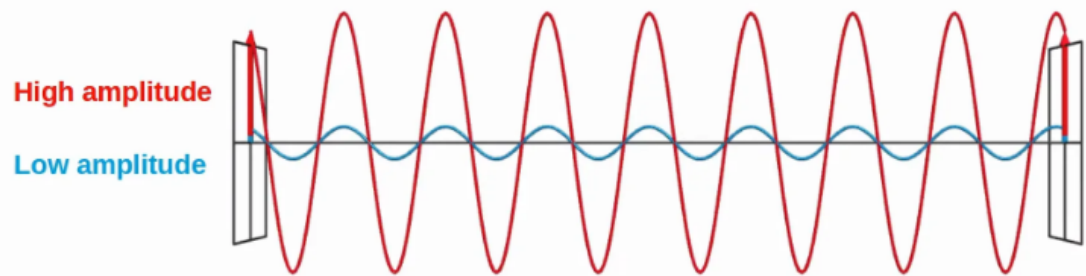
- It means the maximum strength of the electric and magnetic fields.

2. Frequency

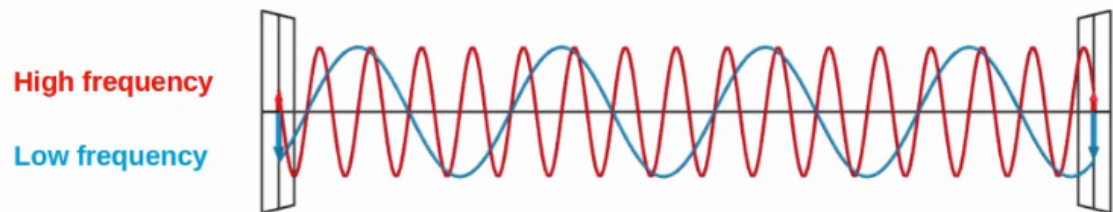
- It measures the number of up/down cycles per a given unit of time.
- The most common measurement of **frequency** is **hertz**.
 - Hz (Hertz) = cycles per second
 - KHz (Kilohertz) = 1,000 cycles per second
 - MHz (Megahertz) = 1,000,000 cycles per second
 - GHz (Gigahertz) = 1,000,000,000 cycles per second
 - THz (Terahertz) = 1,000,000,000,000 cycles per second

3. Period

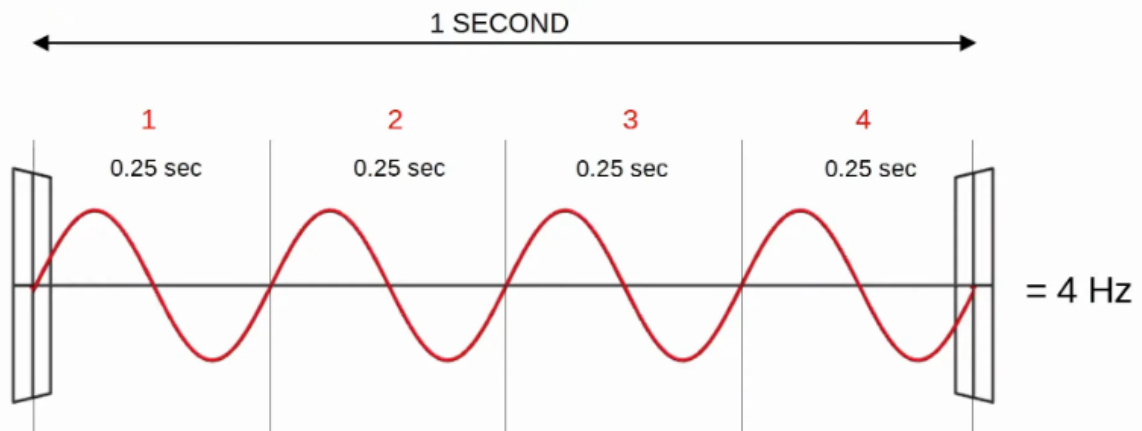
- It represents the amount of time for one cycle to be done.



Amplitude Demonstration



Frequency Demonstration



Period Demonstration

Wireless signals use radio frequency ranges in the *Ultra high frequency* and *Super high frequency* range.

Radio Frequency - Bands

Wi-Fi uses two main bands (frequency ranges):

1. **2.4 GHz** band
2. **5 GHz** band

Band information

- The **2.4 GHz** band typically provides further reach in open space and better penetration of obstacles such as walls.
- However, more devices tend to use the **2.4 GHz** band so interference can be a bigger problem compared to the **5 GHz** band.
- **Wi-Fi 6 (802.11ax)** has expanded the spectrum to include a band in the **6 GHz** range.

Radio Frequency - Channels

Each band is divided up into multiple 'channels'.

- Devices are configured to transmit and receive traffic on one (or more) of these channels.
- In small wireless LANs with only a single **AP**, you can use any channel.
- However, in larger WLANs with multiple **APs**, it's important that adjacent **APs** don't use overlapping channels in order to avoid interference.
- The **2.4 GHz** band is split up into multiple channels
 - In this band, it is recommended to use channels **1, 6, 11**.
 - When using channels **1, 6, 11**, you can place **APs** in a '*honeycomb*' pattern to provide complete coverage of an area without interference between channels.
- The **5 GHz** band consists of non-overlapping channels, so it is much easier to avoid interference between adjacent **APs**.

Radio Frequencies - 802.11 Standards

Standard	Frequencies	Max Data Rate	Alternate Name
802.11	2.4 GHz	2 Mbps	---
802.11b	2.4 GHz	11 Mbps	---
802.11a	5 GHz	54 Mbps	---
802.11g	2.4 GHz	54 Mbps	---
802.11n	2.4 / 5 GHz	600 Mbps	Wi-Fi 4 High Throughput (HT) Wi-Fi
802.11ac	5 GHz	6.93 Gbps	Wi-Fi 5 Very High Throughput (VHT) Wi-Fi
802.11ax	2.4 / 5 / 6 GHz	4 x 802.11ac (4 x 6.93)	Wi-Fi 6

Service Sets

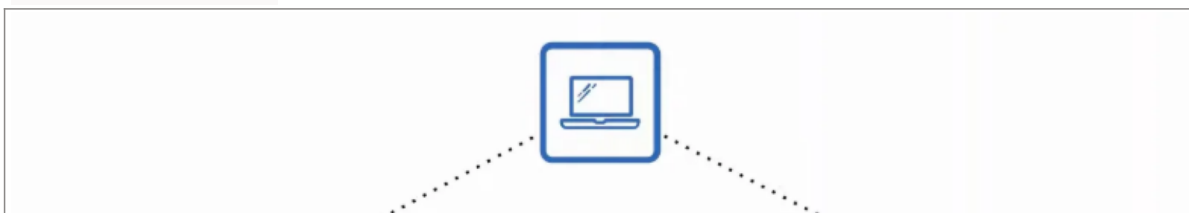
802.11 defines different kinds of service sets which are groups of wireless network devices.

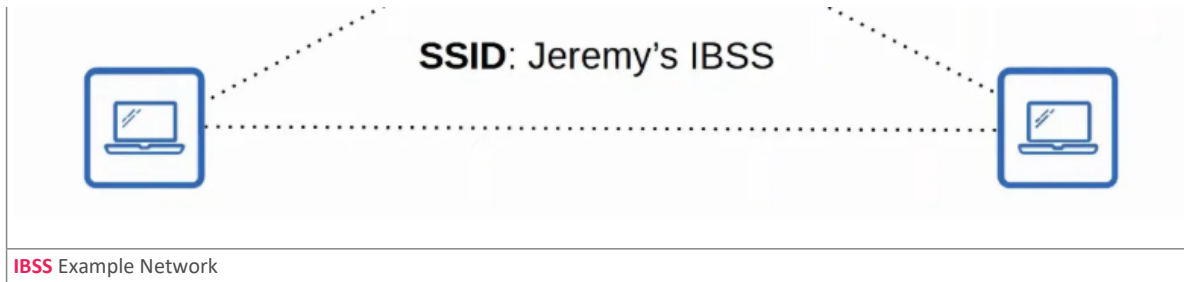
- There are three main types:
 1. **Independent**
 2. **Infrastructure**
 3. **Mesh**
- All devices in a **service set** share the same **Service Set Identifier (SSID)**.
 - The **SSID** is a human-readable name which identifies the **service set**.
 - The **SSID** does not have to be unique

1. IBSS

An Independent Basic Service Set (IBSS) is a wireless network in which two or more wireless devices connect directly without using an Access Point (AP).

- Also called an **ad-hoc** network.
- Can be used for file transfers.

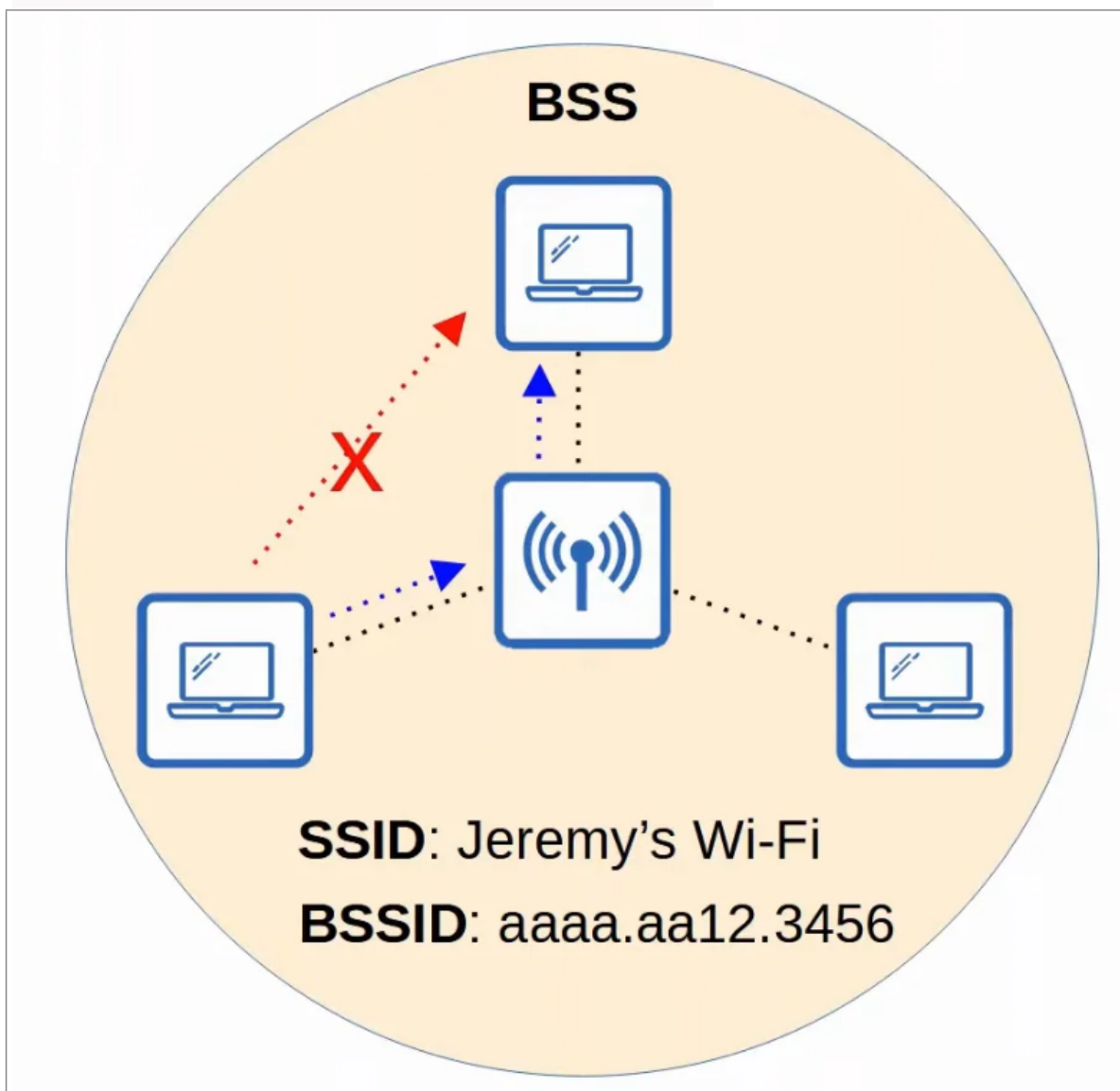




2. BSS

A Basic Service Set (BSS) is a kind of *Infrastructure* Service Set in which clients connect to each other via an AP, but not directly to each other.

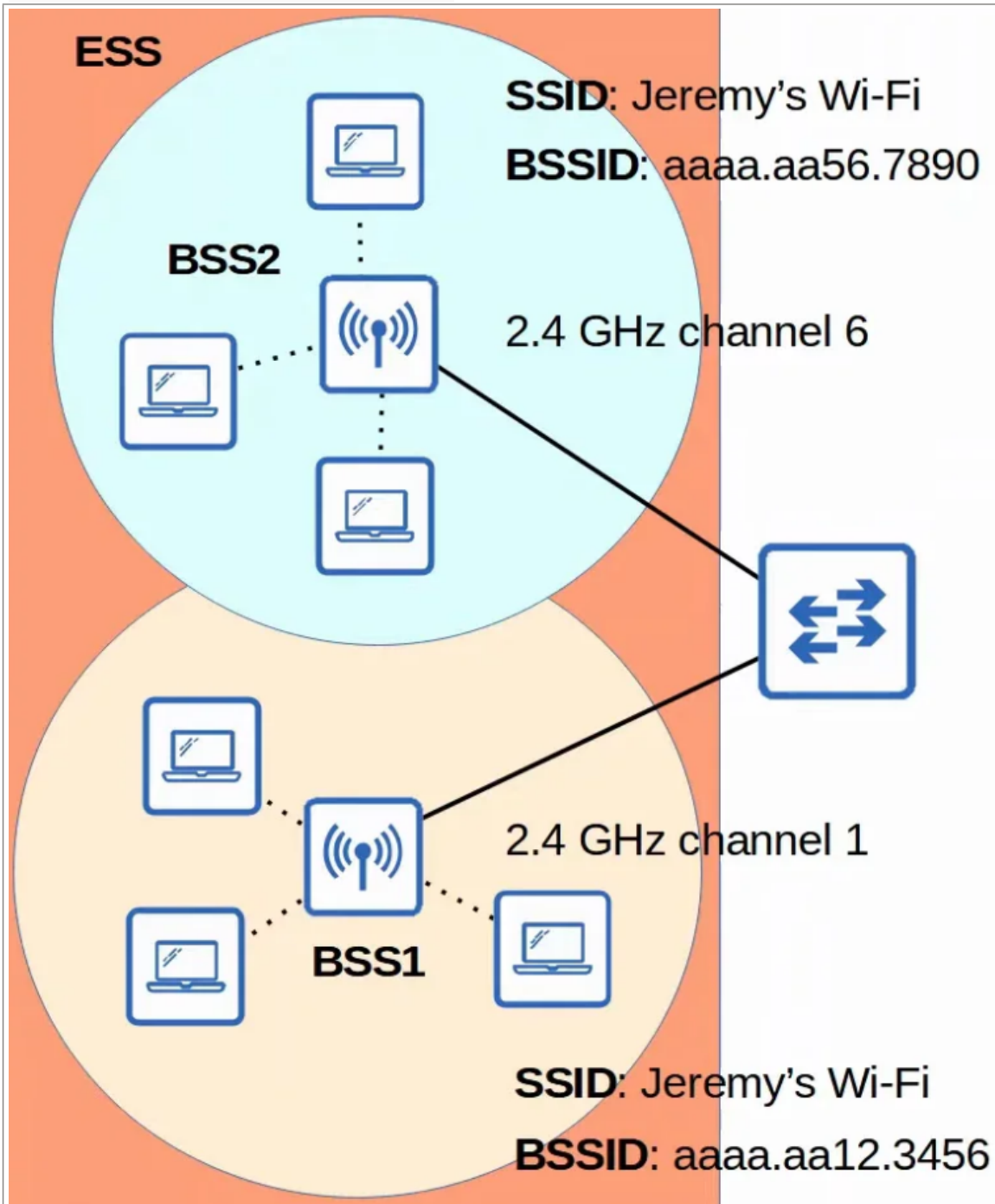
- A **Basic Service Set ID (BSSID)** is used to uniquely identify the **AP**.
 - The **BSSID** is the MAC address of the **AP**'s radio.
 - Other **AP**s can use the same **SSID**, but not the same **BSSID**.
 - The area around an **AP** where its signal is unable is called a **Basic Service Area (BSA)**
 - *(Clients must communicate with each other via the AP, not directly with each other.)*
- SSID vs BSSID**
- **AP**s in a network can use the same **SSID (Service Set Identifier)**
 - However, they cannot use the same **BSSID (Basic Service Set ID)** as it is unique for each device.
- Wireless devices request to associate with the **BSS**.
 - Wireless devices that have associated with the **BSS** are called '*clients*' or '*stations*'



3. ESS

An Extended Service Set (ESS) is used to create larger wireless LANs beyond the range of a AP.

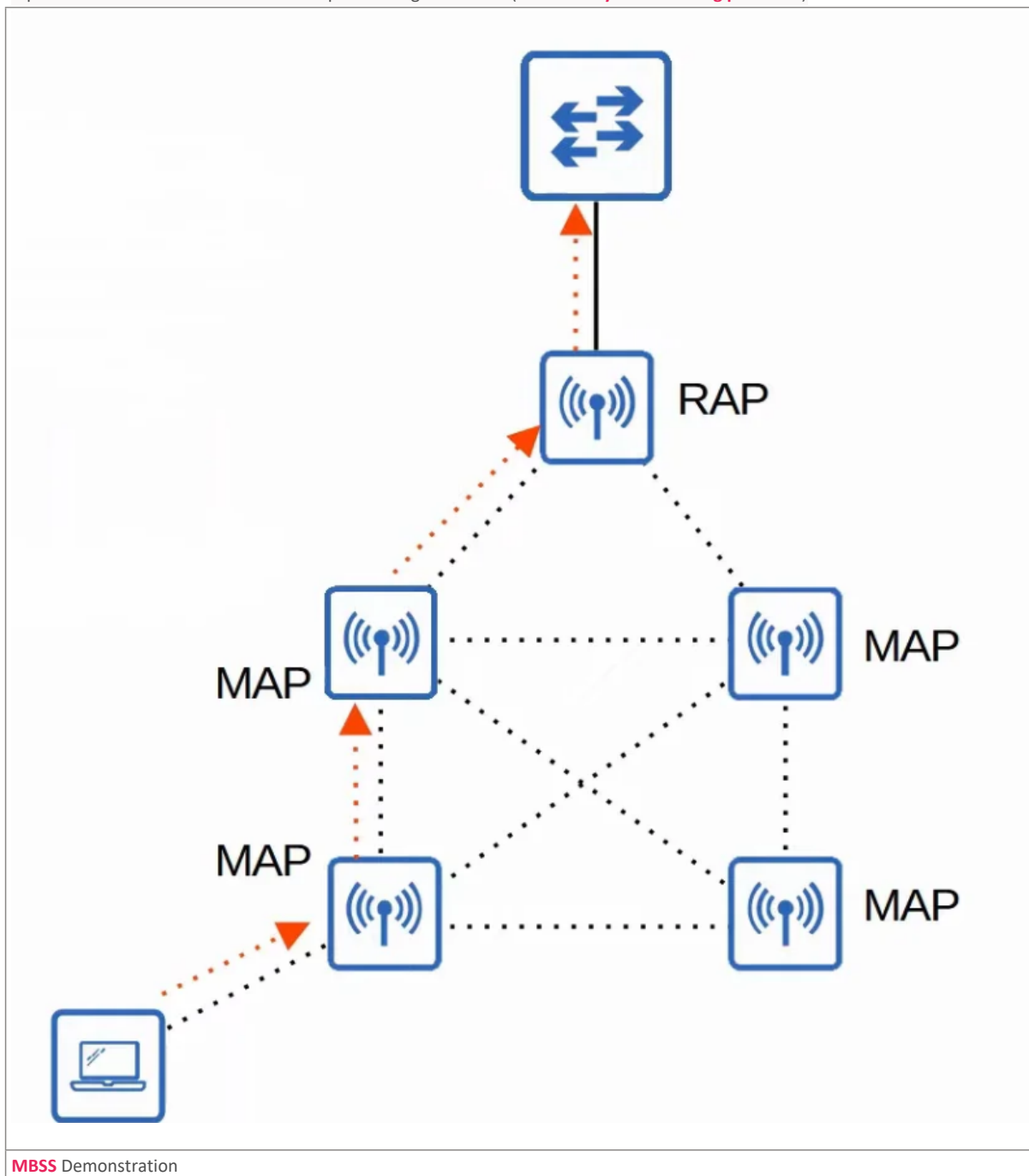
- **APs** with their own **BSSs** are connected by a wired network.
 - Each **BSS** uses the same **SSID**.
 - Each **BSS** has a unique **BSSID**.
 - Each **BSS** uses a different channel to avoid interference.
- **ESS** provides a feature called **roaming**, where clients can pass between **APs** without having to reconnect, providing a seamless **Wi-Fi** experience when moving between **APs**.



4. MBSS

A Mesh Based Service Set (MBSS) can be used in situations where it's difficult to run an Ethernet connection to AP.

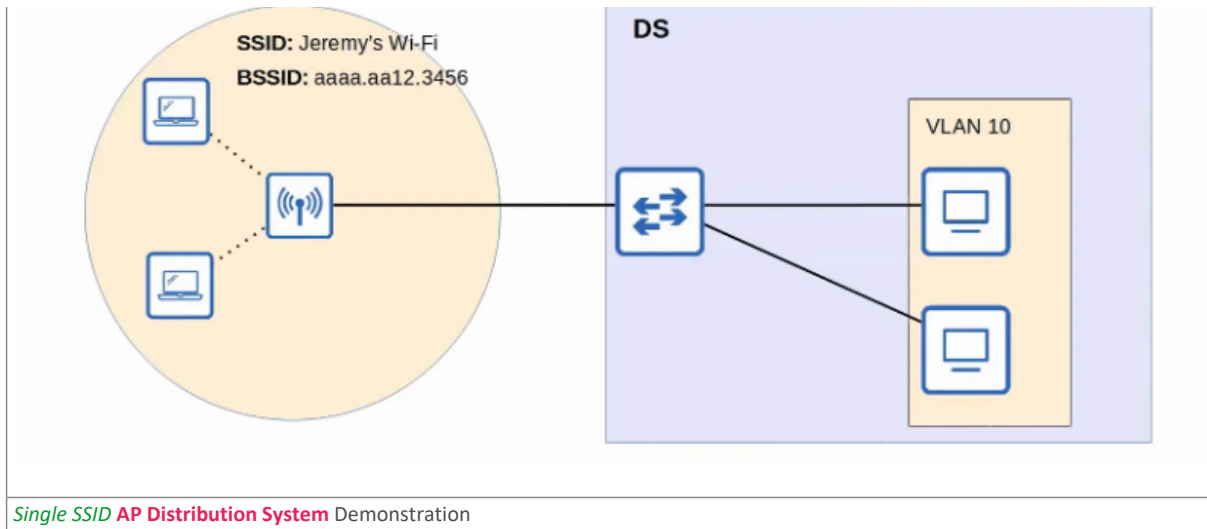
- **Mesh APs** use two radios:
 - One to provide a **BSS** to wireless clients
 - One to form a '*backhaul network*' which is used to bridge traffic from **AP** to **AP**.
- At least one **AP** is connected to the wired network, and it is called the **Root Access Point (RAP)**.
- The other **APs** are called **Mesh Access Points (MAPs)**.
- A protocol is used to determine the best path through the mesh (similar to **dynamic routing protocols**)



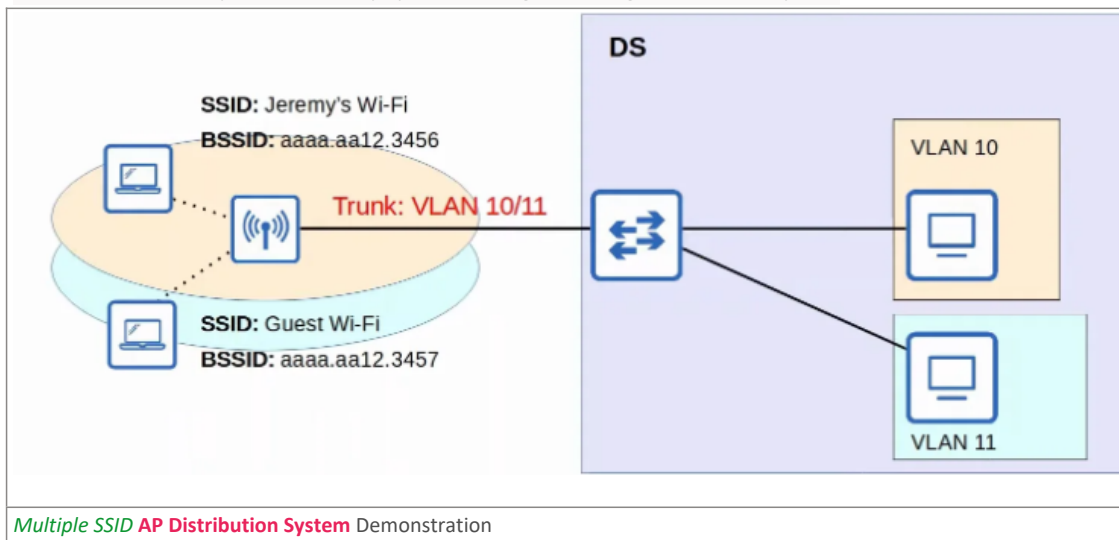
Distribution System

Most wireless networks aren't standalone networks. Rather, they are a way for wireless clients to connect to the *wired* network infrastructure.

- In **802.11**, the upstream *wired* network is called the **Distribution System (DS)**.
- Each wireless **BSS** or **ESS** is mapped to a VLAN in a *wired* network.



- It is also possible for an **AP** to provide multiple wireless LANs, each with a unique **SSID**.
- Each **WLAN** is mapped to a separate VLAN and connected to the wired network via a **trunk**.
- Each **WLAN** uses a unique **BSSID**, usually by incrementing the last digit of the **BSSID** by one.



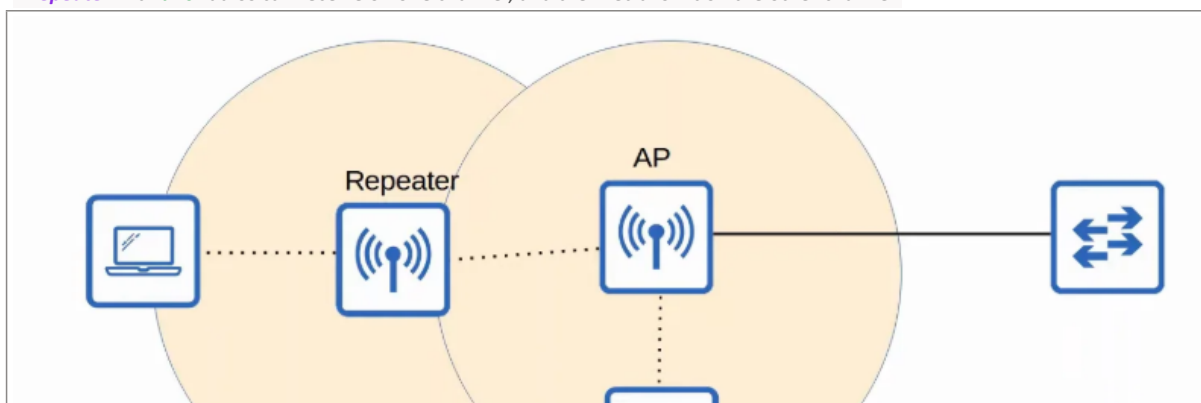
Additional AP Operational Modes

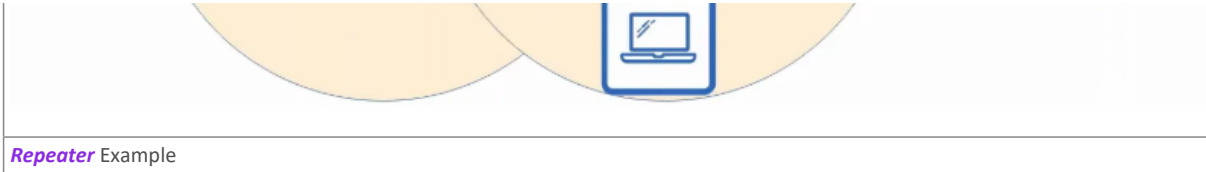
APs can operate in additional modes beyond the Service Sets discussed previously.

1. Repeater

An AP running in **repeater** mode can be used to extend the range of a BSS.

- The **repeater** will simply retransmit any signal it receives from the **AP**.
- A **repeater** with a **single** radio must operate on the same channel as the **AP**, but this can drastically reduce the overall throughput on the channel.
- A **repeater** with **two** radios can receive on one channel, and then retransmit on the other channel.

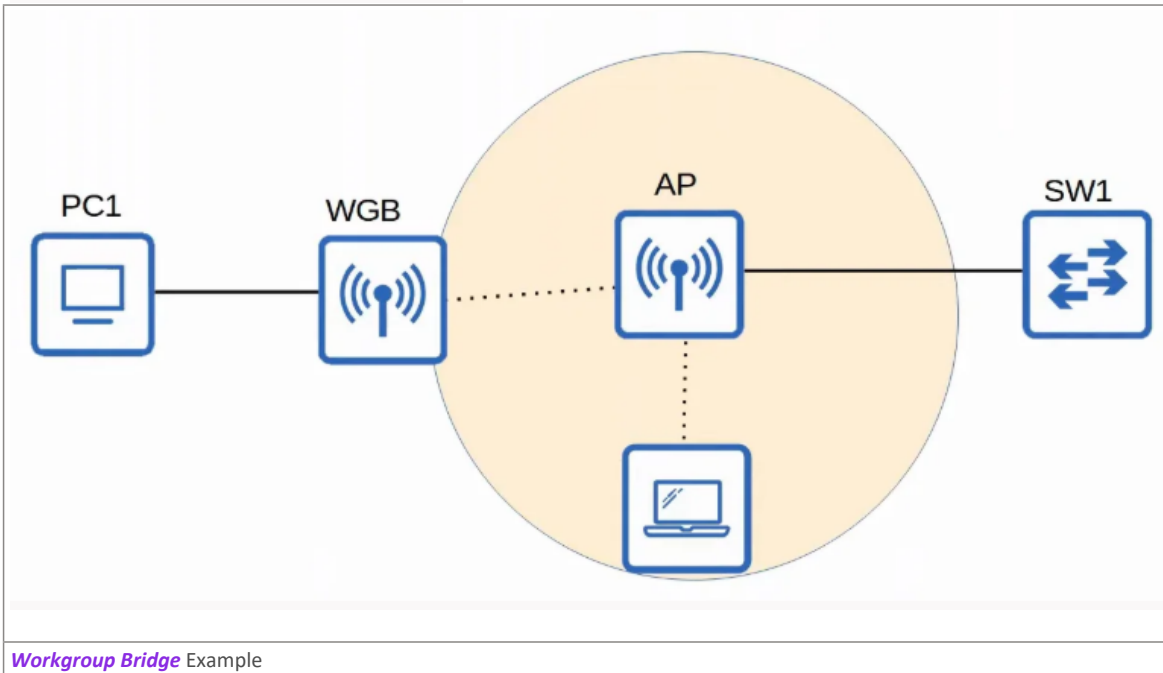




2. Workgroup Bridge (WGB)

A **workgroup bridge (WGB)** operates as a wireless client of another AP, and can be used to connect **wired** devices to the **wireless** network.

- In the example, **PC1** does not have wireless capabilities and does not have access to a **wired** connection like **SW1** for example.
- There is where **workgroup bridges** come in, they allow you to connect **wired** devices (**PC1** here) to the **workgroup bridge** which has a wireless connection to the **AP**.



The two kinds of WGBs

- Universal WGB (uWGB)**
 - It is a **802.11** standard that allows one device to be bridged to the **wireless** network.
- WGB**
 - It is a **Cisco proprietary** version of the **802.11** standard that allows multiple **wired** clients to be bridged to the **wireless** network.

3. Outdoor Bridge

An **outdoor bridge** can be used to connect networks over long distances without a physical cable connecting them.

- The **APs** will use specialized antennas that focus the signal power in one specific direction, which allows the **wireless** connection to be made over longer distances than normally possible.
- The connection can be **point-to-point**, or **point-to-multipoint** in which multiple sites connect to one central site (like the **Hub and Spoke** topology).

